BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Syntax and Logic Errors

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why must a trace table follow executed assignments in order?

2. Why open a file before reading it?
   A) To create a file handle
   B) To delete its contents
   C) To encrypt every line
   D) To print it automatically

3. "Reusing one short password everywhere improves account security." Correct this. Explain why.

4. A file is split into fragments on an HDD. "Defragmentation speeds up every SSD by moving its read head." Correct this. When does defragmentation help?

Today's objectives

Distinguish syntax errors from logic errors. Use a trace to identify a fault.

Key vocabulary - Read, cover, recall

Syntax error
Statement breaking the grammar of a language.

Logic error
Program runs but produces a wrong result.

Trace table
A record of variable values as an algorithm executes.

Debugging
Finding and correcting program faults.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Syntax errors

A misspelled IF keyword or missing ENDIF violates grammar. In a runnable implementation it may prevent the code from executing.

Logic errors

A program may run but give the wrong result, for example using addition when the task needs multiplication.

Find the first incorrect step

Trace variables as instructions execute. Compare expected and observed values to pinpoint the first incorrect result.

Fix and retest

Correct the faulty instruction then repeat the failed case and nearby boundary cases to check the effect.

I Do - Worked model

A ticketing program calculates total = price + quantity when it should calculate cost for multiple tickets. With price = 4 and quantity = 3, identify the error, correct it and explain the result. [4 marks]

- The expected cost is three tickets priced at £4 = £12 [1].
- The program uses addition, producing 4 + 3 = 7 [1].
- Replace + with multiplication * [1].
- After the correction 4 * 3 gives 12 [1].

We Do - Guided practice

1. A sports club program should find four times a membership fee, but adds 4 to it. With fee = 6, explain the logic error and fix. [4 marks]

2. An IF block is missing ENDIF in an example. Explain the syntax error and how to correct it. [2 marks]

3. A pupil corrects a program but tests it only once. Explain two improvements to the testing process. [4 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Fix the calculation that converts 32 bits into bytes.

```
bits = 32
bytes = bits * 8
print(bytes)
```

You Do - Independent exam questions

Q2. State the difference between a syntax error and a logic error. [2 marks]

Q3. A fee calculator adds 5 to the number of tickets instead of multiplying by 5. With 3 tickets, identify the error and correct the calculation. [4 marks]

Q4. Explain how tracing and retesting can identify, fix and confirm both syntax and logic errors in a program. [6 marks]

Exit Ticket - True or False?

A logic error can cause an incorrect output even when the program runs.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Each variable value depends on the state produced by earlier executed instructions.

2. A - To create a file handle

3. They make guessing harder and prevent reuse of a compromised password across accounts.

4. Defragmentation brings file fragments together on an HDD, reducing read-head movement. It rearranges data; it does not encrypt or compress it.

I Do - Explain the model

The expected cost is three tickets priced at £4 = £12 [1].

The program uses addition, producing 4 + 3 = 7 [1].

Replace + with multiplication * [1].

After the correction 4 * 3 gives 12 [1].

We Do - Model answers

1. Addition would calculate 6+4=10 [1], rather than 24 [1]. Change operator to multiplication [1], producing 6*4=24 [1].

2. Missing ENDIF leaves an IF block unclosed [1]. Add ENDIF after the appropriate branch to complete the block [1].

3. Repeat the originally failing test [1] to confirm the bug is fixed [1]. Also test a boundary or another input [1] to check the change did not cause another failure [1].

Code First - Answer

```
bits = 32
bytes = bits DIV 8
print(bytes)
```

Expected output: 4

You Do - Self-mark

Q2: Syntax: invalid language grammar [1]; logic: program runs but produces incorrect results [1].

Q3: Adding gives 8 [1] but expected price is 15 [1]; use multiplication [1], giving 3*5=15 [1].

Q4: A syntax error breaks language rules [1] and should be corrected according to diagnostic feedback [1]. Logic errors produce wrong results [1] and trace tables show the first incorrect value [1]. Repeat failed test after correction [1] and test another normal/boundary case to avoid regressions [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.